



Data & AI Governance Capabilities

Governance is about standards, policies and controls.

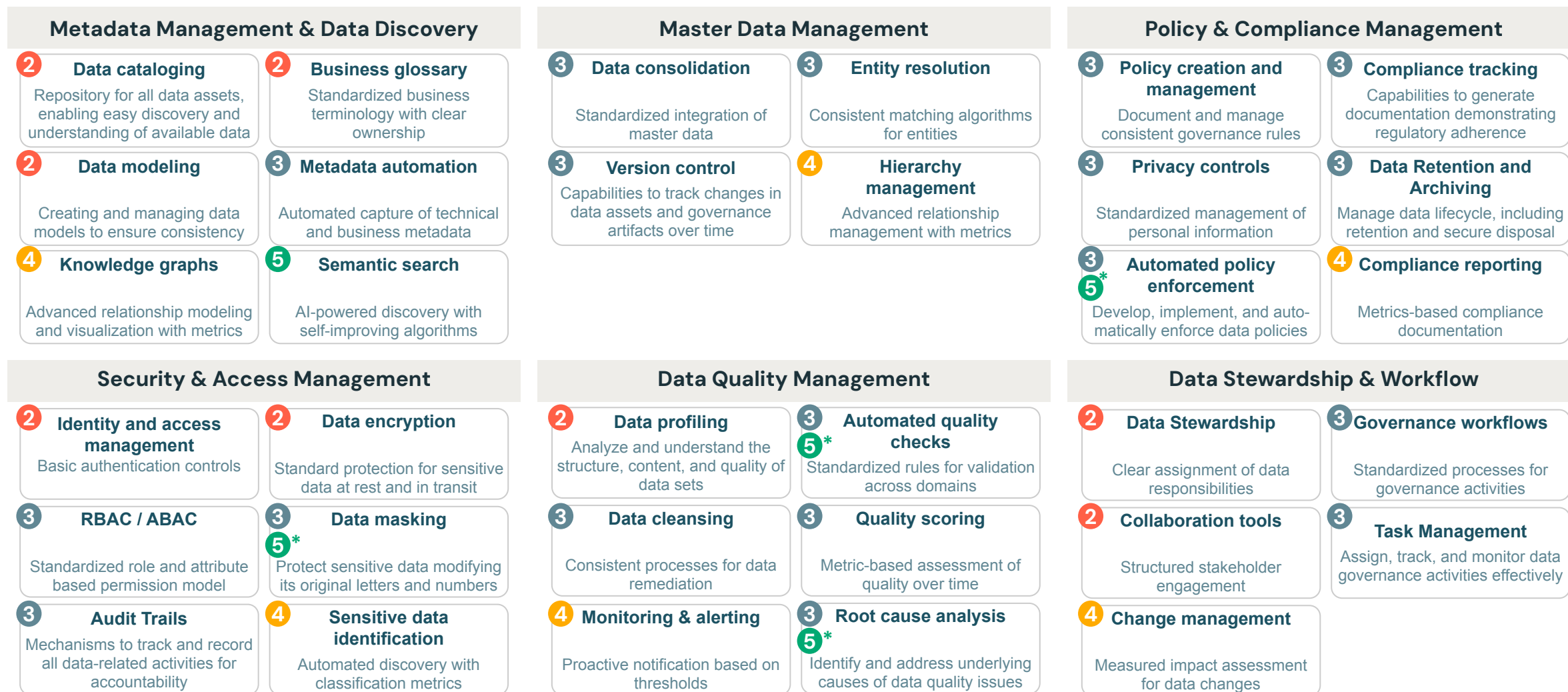
For platforms to integrate into a governance operating model, they need to provide the **right set of capabilities**.

There are different sets of **capabilities for data and for AI governance**, but they **work hand in hand**.

The **governance maturity determines** what needs to be implemented and what shouldn't be implemented yet

And Databricks unified governance provides many capabilities in Unity Catalog and other parts of the platform

Data Governance Capabilities (1/2)



Data Governance Capabilities (2/2)

Data Lineage & Impact Analysis

- 3 Automated end-to-end lineage capture**
Comprehensive tracking from source to consumption
- 3 Visualization tools**
Standard representation of data flows
- 4 Impact analysis**
Assess how changes in data or policies affect downstream processes and applications
- 3 Root cause investigation**
5* Tracing of data related issues

Measurement & Optimization

- 4 Governance metrics**
Comprehensive KPIs for governance effectiveness
- 4 Value tracking**
Quantified business impact measurement
- 4 Customizable dashboards**
Visualize and monitor key data governance metrics and KPIs
- 4 Maturity assessment**
Benchmark current state and identify areas for improvement in data governance
- 5 Predictive Analytics**
Anticipate potential data governance issues and take proactive measures
- 5 Continuous improvement**
Self-optimizing governance processes

Integration & Interoperability

- 2 API connectivity**
Basic integration interfaces
- 2 Data Sharing**
Tools to facilitate secure data sharing within and outside the organization
- 3 Cross-platform governance**
Consistent governance across environments (e.g. clouds)
- 3 Metadata exchange**
Secure and consistent metadata sharing between systems

AI & Automation Support

- 5 AI-driven metadata management**
AI-driven automated metadata capture and enrichment
- 5 Algorithm transparency**
Advanced explainability capabilities
- 5 Intelligent Data Quality**
AI-driven automated data quality checks and improvements
- 5 Anomaly detection**
Predictive identification of governance issues



AI Governance Capabilities (1/2)



AI Governance Capabilities (2/2)

Risk Management & Assessment

- 2 Model Validation and Testing**
Capabilities to rigorously test AI models before deployment
- 3 Risk identification**
Automated detection of potential risks in AI systems
- 3 AI risk impact analysis**
Tools to determine potential business and ethical impacts of AI deployments
- 3 AI Incident Response**
Protocols and tools for managing and mitigating AI-related incidents
- 4 Risk scoring**
Quantitative evaluation of risk levels for prioritization

Transparency & Explainability

- 3 Explainability tools**
Capabilities to make AI decision-making understandable
- 3 Interpretability visualizations**
Visual representations of how AI models arrive at conclusions
- 3 Audit Trail for AI Decisions**
Track and record AI-driven decisions for accountability
- 4 Decision pathway mapping**
Tracing of factors influencing AI outputs
- 5 Technical documentation automation**
Generation of model explanation artifacts
- 5 Stakeholder-appropriate explanations**
Self-improving explanations by audience

AI Education and Training

- 3 AI Literacy Programs**
Tools and resources to improve AI understanding across the organization
- 3 AI Ethics Training**
Platforms for delivering and tracking completion of AI ethics education
- 3 AI Skills Assessment**
Capabilities to evaluate and monitor AI-related skills within the organization

Fairness & Ethical AI

- 2 Bias detection and mitigation**
Tools to identify/address potential biases in models, data
- 2 Fairness monitoring**
Quantitative measures of model fairness across different demographics
- 3 Ethical guidelines implementation**
Frameworks for embedding ethical principles in AI
- 3 Ethical AI Assessment**
Tools to evaluate AI systems for alignment with ethical principles and organizational values
- 4 Fairness monitoring**
Quantitative measures of model fairness across different demographics

Compliance & Regulatory Management

- 2 Compliance assessment**
Evaluation of AI systems against regulatory requirements
- 3 Compliance documentation**
Generation of evidence for regulatory submissions
- 3 Compliance reporting**
Capabilities to generate reports demonstrating adherence to AI-specific regulations
- 4 AI Regulatory Tracking**
Tools to monitor and interpret evolving AI regulations globally
- 5 Automated policy enforcement**
Automated implementation of regulatory requirements

Stakeholder Engagement & Impact Assessment

- 2 Stakeholder Feedback Mechanisms**
Tools for gathering & analyzing feedback on AI systems
- 4 Public Trust Monitoring**
Tools to gauge and track public perception and trust in the organization's AI initiatives
- 4 AI Impact Assessment**
Frameworks for evaluating the societal and environmental impact of AI systems



Data & AI Governance Operational Model

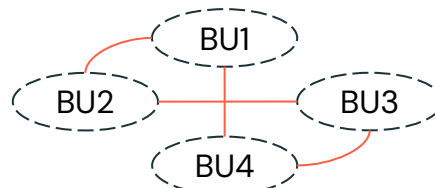
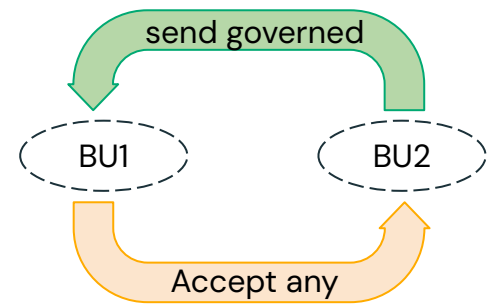
There is not just one way to drive Data & AI Governance.

The company structure demands for different
operating models for governance

Widely followed Laws & Principles

Organizations, often unintentionally, follow these laws & principles in absence of a proper governance operational models

 Business Unit (BU)

Approach	Description		Limitations
Conway's Law (wikipedia)	Named after computer scientist and programmer Melvin Conway, and it states <i>"Organizations which design systems are constrained to produce designs which are copies of the communication structures of these organizations"</i> . <i>In other words, everyone will do their own thing.</i>		This can lead to significant issues • Weak communication and coordination • Siloed data teams and fragmented data practices • Misalignment of incentives and ownership • Inflexibility and lack of resilience
Robustness Principle (wikipedia)	A design guideline for software that states: <i>"be conservative in what you send, be liberal in what you accept from others"</i> . That is when multiple business units (BU) manages their own data and AI, BUs tries to be liberal in accepting data from other BUs even in different schemas/formats, however be strict when they share data & AI assets by following a governed schemas/formats. <i>In other words, BUs will still manage to communicate even with no governance Operating Model in place.</i>		Security Risks Can be exploited, malicious actors exploit tolerant systems, making strict input validation essential to prevent attacks. Interoperability Issues Lenient acceptance creates a "one-way ratchet": consumers must adapt to producers' deviations, entrenching bugs as de facto standards.

Governance Operational Model

There is no one-size-fits-all for data and AI governance

The organizational structure for implementing governance. There are four primary variants:

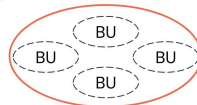
- **Centralized**
 - Single governing body
 - Highly regulated industries
- **Decentralized**
 - Fully distributed with minimal central oversight
 - Agile organizations prioritizing innovation and rapid iteration
- **Federated**
 - Distributed ownership with strong central oversight
 - Multinational corporations
- **Hybrid**
 - Mix of centralized and decentralized
 - Post-merger organizations

Operational Governance Model (1/2)

Governance models can adopt different structures

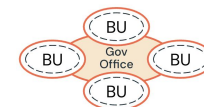
Centralized Governance

- A central governing body or team is responsible for setting and enforcing data and AI policies, standards, and best practices, ensuring uniformity and consistency organization-wide.
- Suits organizations operating in highly regulated environments or those prioritizing consistency.
- The centralization of decision-making may limit flexibility, adaptability, and responsiveness to local needs.



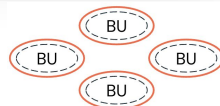
Federated Governance

- Combines centralized oversight with decentralized execution. Business units govern their data, while a central body ensures alignment and consistency.
- Suited for multinational corporations or conglomerates with diverse business units.
- Strengths: Balances autonomy with collaboration, leveraging localized expertise



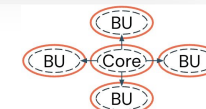
Decentralized Governance

- Allows individual departments to independently manage and adapt data and AI practices according to their operational needs while ideally aligning with broader organizational goals.
- Provides flexibility and enables departments to apply specialized knowledge, tailoring data and AI solutions to unique contexts.
- The decentralized structure can lead to duplication of skills, efforts and technology, and variability in risk management, ethical standards, and regulatory compliance across units.



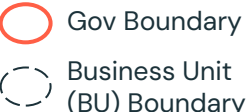
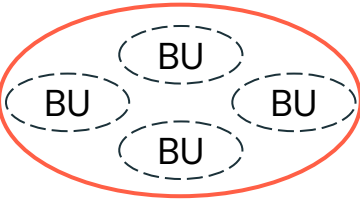
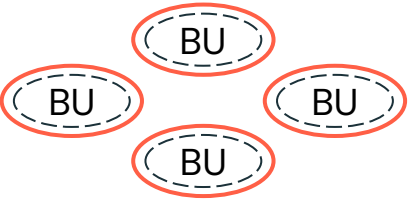
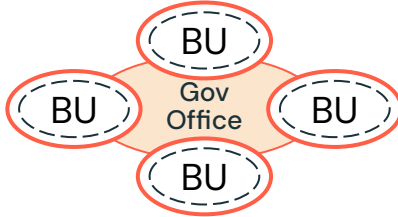
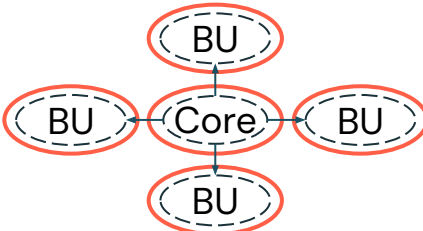
Hybrid Governance

- Combines centralized and decentralized governance elements and balances uniform standards with local flexibility.
- Typically managed by a Center of Excellence, hybrid governance provides centralized oversight while allowing departments to leverage localized knowledge and innovation.
- While effective for organizations with complex data and AI programs or those undergoing transformations, hybrid governance requires careful coordination to avoid overlapping roles or inefficient resource use.



Governance Operational Models (2/2)

Different types of Operational Models and some of their characteristics

	Centralized Governance	Decentralized Governance	Federated Governance	Hybrid Governance
				
Definition	Single authority controls data & AI policies	Each BU manages its own data & AI policies (follows Conway's Laws)	Central coordination with local autonomy	Explicit split (core vs. BU data & AI)
Decision-Making	Centralized, top-down	Distributed, independent	Federated – central guidelines with local execution	Tiered (central dictates core rules, BUs their own rules)
Data Ownership	Owned by a central team (central governance)	Owned by individual teams or BU (shared nothing)	Distributed ownership across entities (shared governance)	Core data owned centrally; BU data owned decentralized (shared core data)
Scalability	Limited by central team's capacity	Highly scalable, but may lack coordination	Scalable with structured coordination	Moderate (core data centralized; BUs scale independently)
Compliance & Security	Strong enforcement, high consistency	Varies by team, potential security inconsistency	Central standards with local enforcement	Strict compliance for core data; flexible for domains
Operational Efficiency	Efficient but bureaucratic	Agile but may duplicate efforts and introduce silos	Balanced – efficient yet flexible, but high federation might introduce silos	Potential bottlenecks for core data; efficient elsewhere
Use Case Suitability	Highly regulated industries (finance, healthcare)	Agile businesses, startups, innovation-driven sectors	Large enterprises, multinational corporations	Mixed compliance/innovation needs (e.g., fintech)

Centralised Governance

(Multi-Tenant Lakehouse)

Each BU/domain operates in a different business boundary, however all BUs/domains fall under the same operational boundary that governs all data & AI assets (multi-tenancy)

Pros:

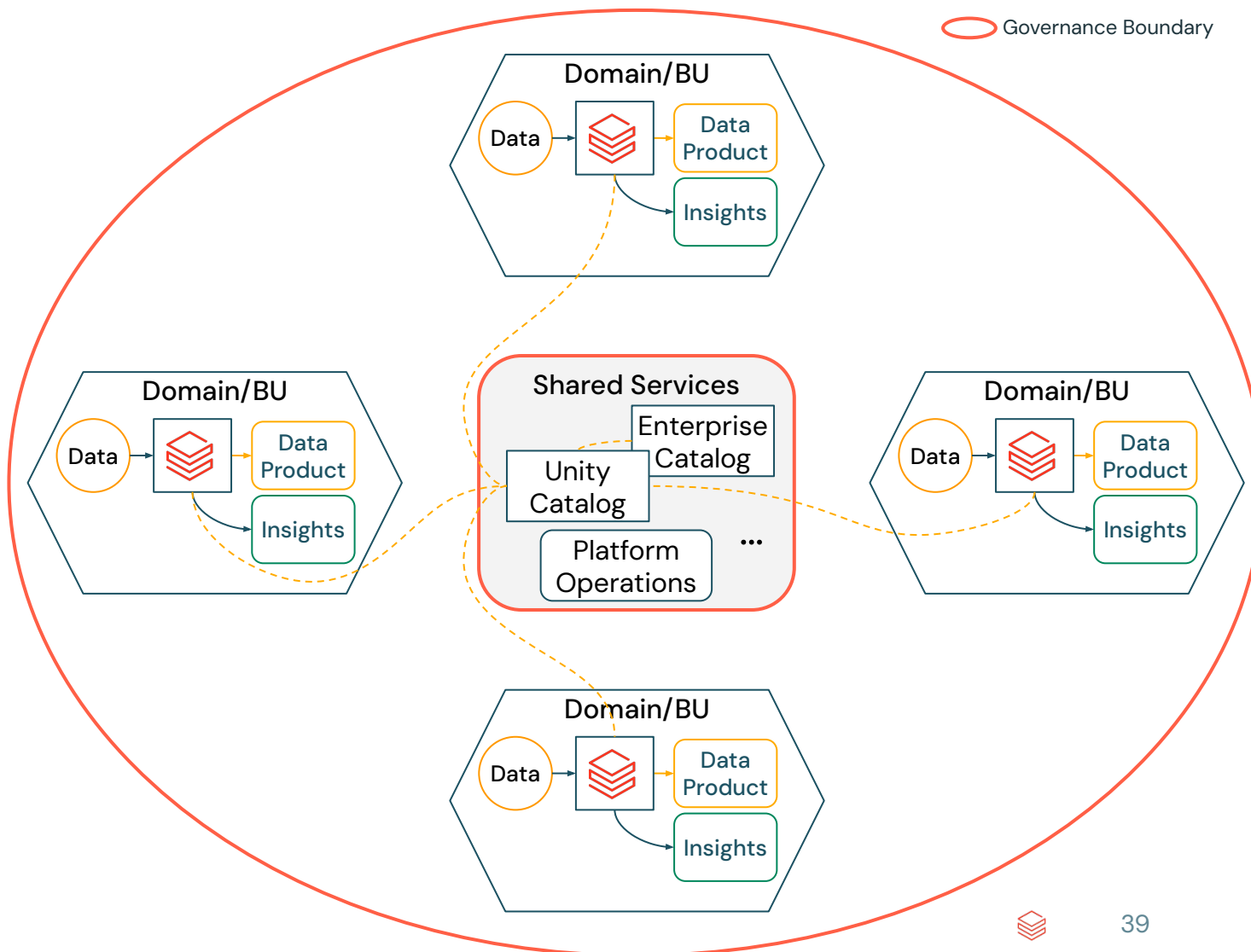
- Strong governance across all BUs/domains
- Unified view of all data and AI assets across the group

Cons

- Might be slow for innovations and decisions making
- Changes are not easy as they have to be agreed across all BUs/domains

Use When:

- Corporate business model is pretty much the same for all BUs/domains
- Operating model is leaning towards centralization (with a centralised IT)
- Strong inter-BU/domain data sharing requirements



Decentralised Governance

(Single Tenant Lakehouses)

Each BU/domain has an independent business and operational boundary (i.e own tenant). Governance is delegated to each BU/domain to define and enforce their own policies

Pros:

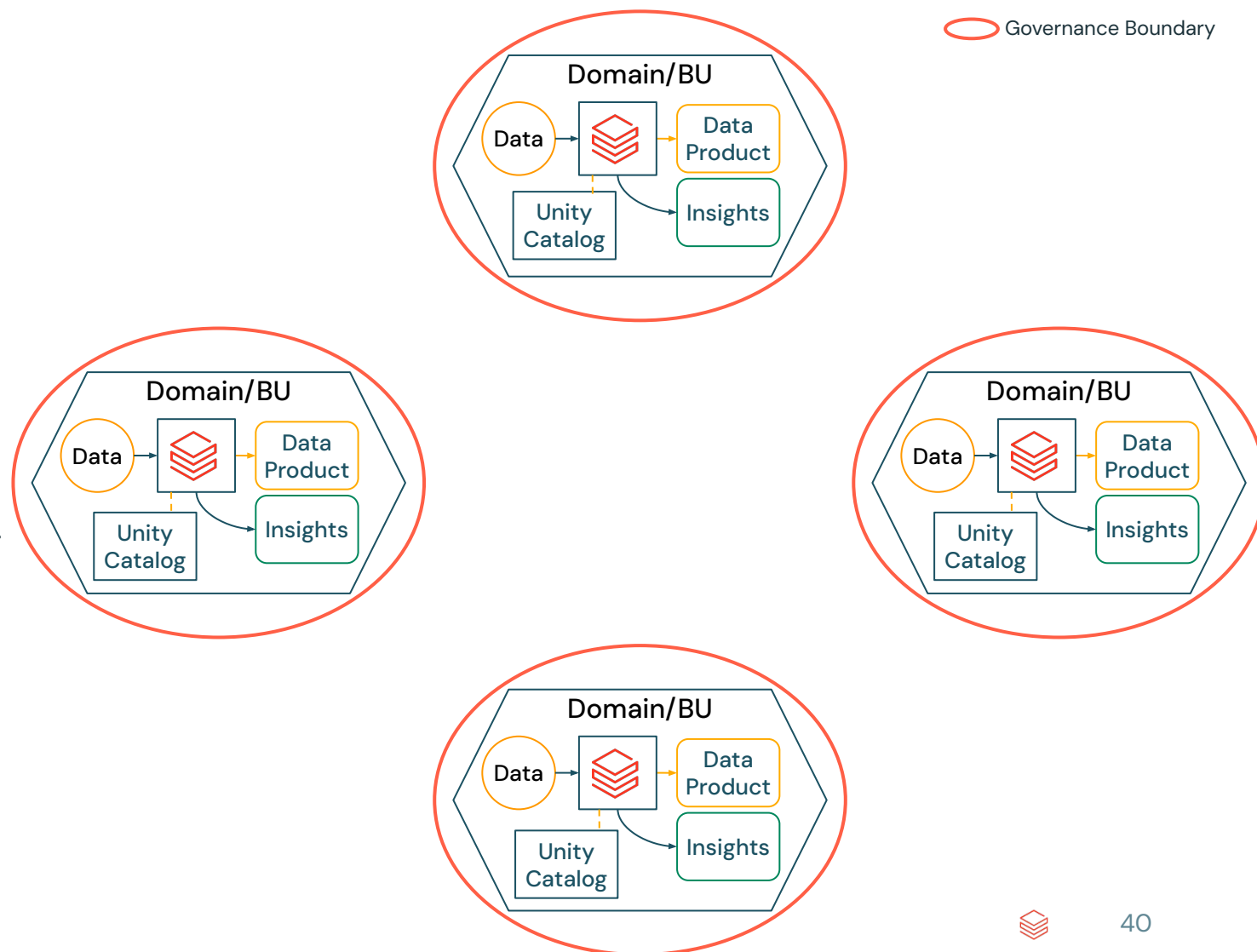
- Greater autonomy for each BU/domain to decide their own data & AI management and strategic goals
- Speed up of innovations and decision making

Cons

- Inconsistent governance perspective
- No strict rules on how each BU/domain defines or maintains their data & AI assets

Use When:

- BUs/domains have different business models independent from one another
- Minimal to none sharing or exchanging information between BUs/domains expected



Federated Governance

(Harmonised Lakehouse Mesh)

Each BU/domain has an independent business and operational boundary. However, governance policies are defined by a centralised Governance-Office, enforced by each BUs – critical assets are governed by the governance office (e.g. shared data products)

Pros:

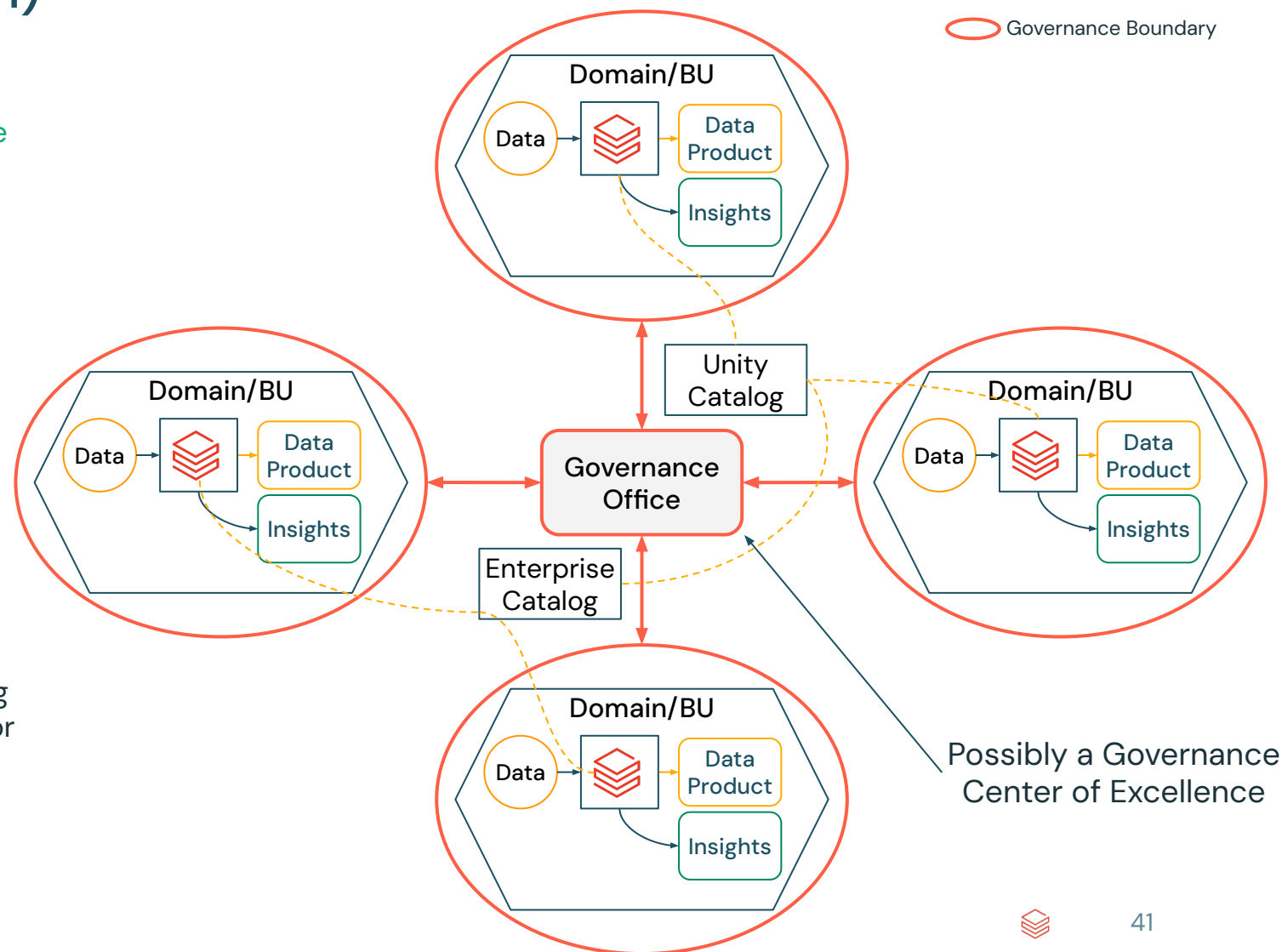
- Speed up of innovation and decision making, yet still maintain centralised governance guidelines
- Possible use of Center of Excellence

Cons

- Hard to enforce governance guidelines across all BUs/domains, without strong governance operations/teams (Governance office)

Use When:

- The company operates each BU/domain independently, however, still requires data sharing between BUs/domains to deliver business goals or comply with regulatory requirements



Hybrid Governance

(Hub & Spoke Lakehouse Mesh)

Midway between centralised and decentralised, allows each BU/domain to govern their own data (decentralised). However, critical business data is collected and governed in one place for all to use (centralised)

Pros:

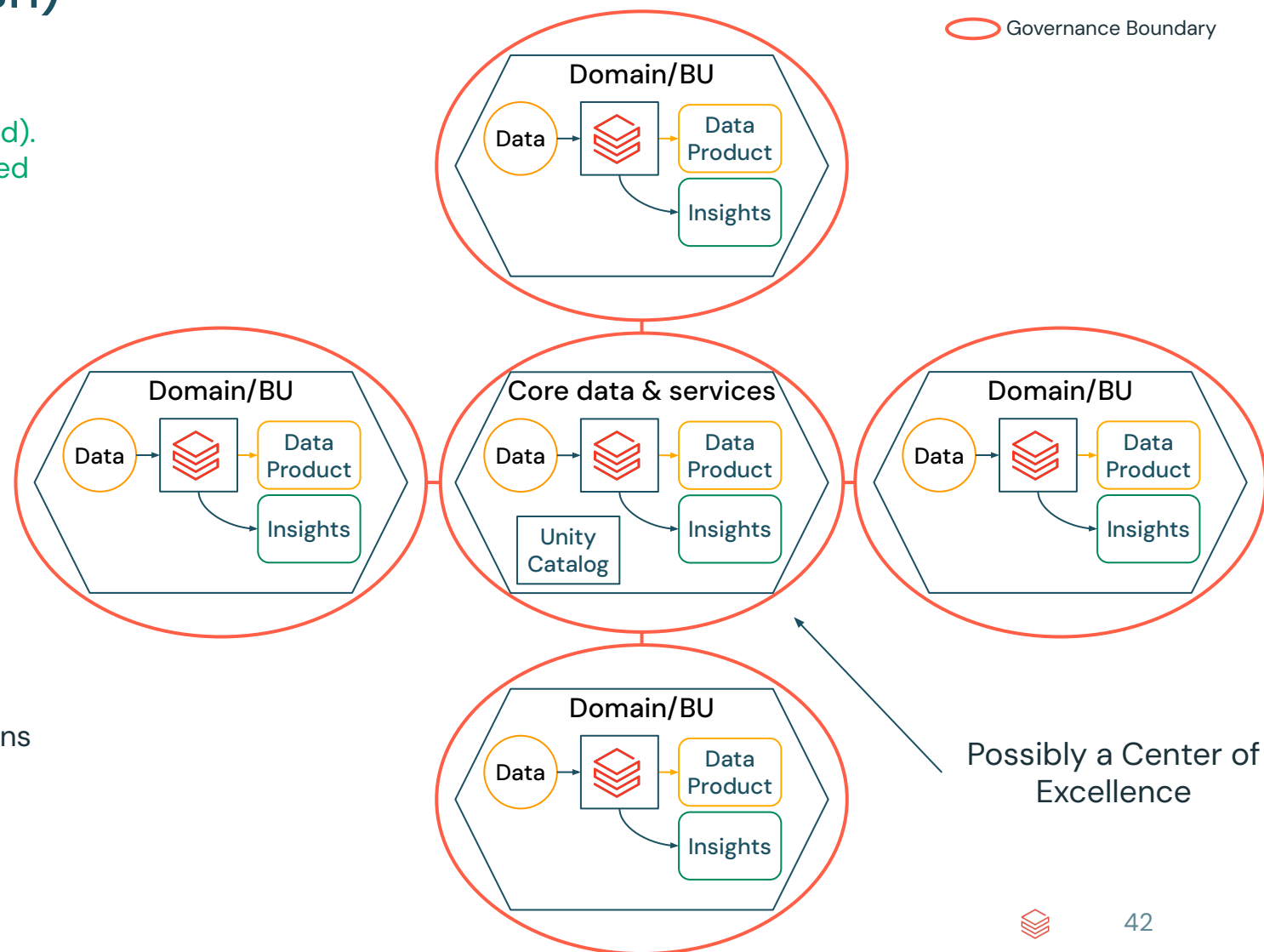
- Speed up of innovations and decision making at the BU/domain level
- Strong governance for critical data that are centrally managed (eg Group Financial details, customer PII data, etc)
- Possible use of Center of Excellence

Cons

- Core data & services will almost act as an independent BU/domain, requires own management, operations and funding

Use When:

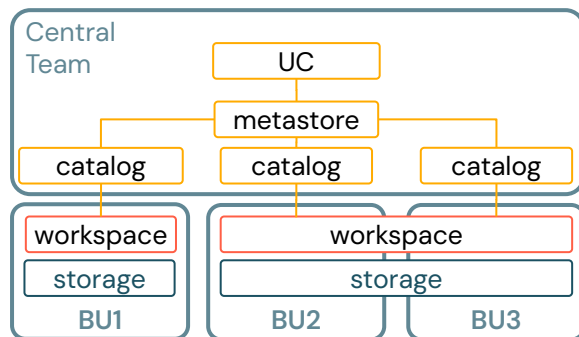
- The company doesn't want to restrict BUs/domains from pursuing own innovation, however, critical business data such as PII/PCI must be centrally managed to comply with regulations



Operational Model impacts UC and Databricks setup

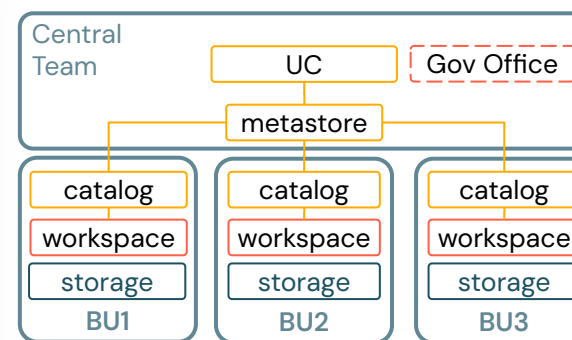
Single Cloud and Single Region

Centralized Governance



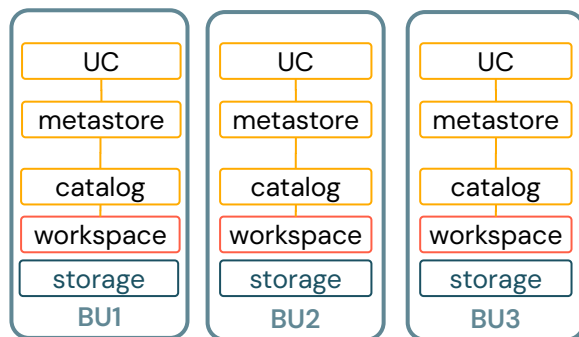
- Central team owns UC down to the catalog level (and sometimes schema)
- It dictates structure and governs data and AI
- BUs own their workspaces and storage (can be shared between BUs)
- BUs can share workspaces and storage accounts

Federated Governance



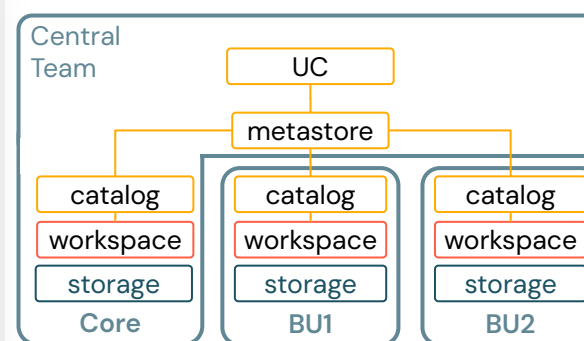
- Central team owns UC down to metastore level (Governance oversight via governance team and metastore admin)
- Central teams has no data
- Independent BUs owning their catalogs

Decentralized Governance



- BUs use different cloud tenants and hence have separate Unity Catalog instances and assigned metastore
- They have their own workspaces and storage accounts

Hybrid Governance

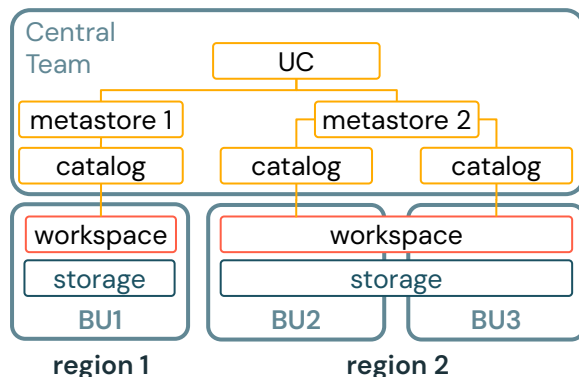


- Central team owns UC, and can own metastore (technical ownership, no metastore admin)
- Central teams owns core (common) data
- BUs have their own catalogs and can be completely independent (own workspaces and storage accounts)
- BUs use core data

Operational Model impacts UC and Databricks setup

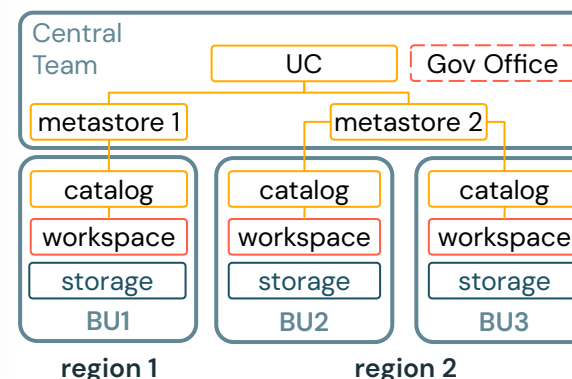
Single Cloud and Multi-region

Centralized Governance



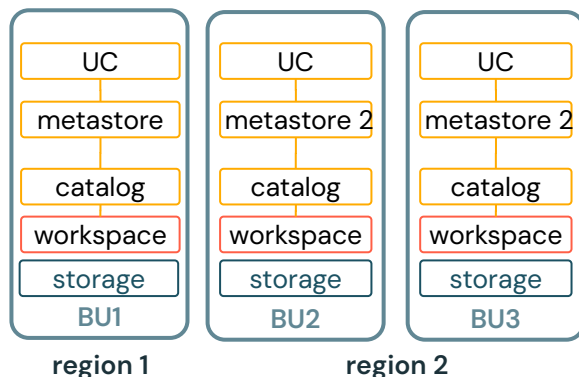
Same as single region, plus
– Central team owns UC down to the catalog level across all regions/metastores

Federated Governance



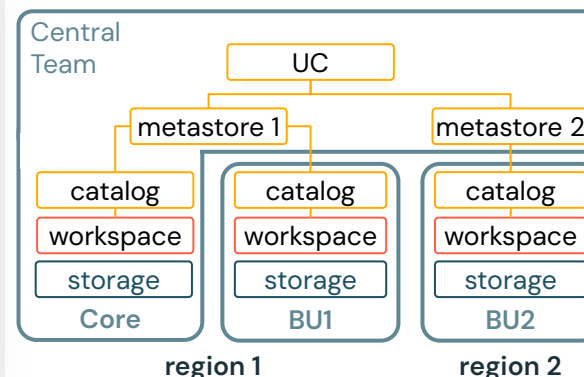
Same as single region, plus
– Central team owns UC down to the level of each metastore (oversight via metastore admin)

Decentralized Governance



Same as single region

Hybrid Governance

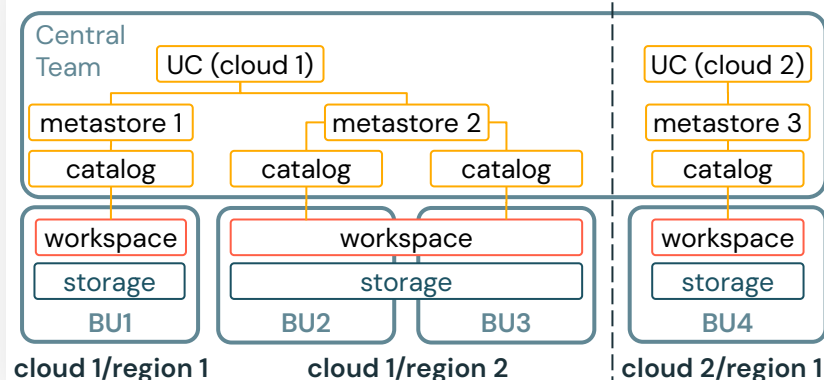


Same as single region, plus
– Central team owns UC, and can own one or more metastores of the regions (technical ownership, no metastore admin)

Operational Model impacts UC and Databricks setup

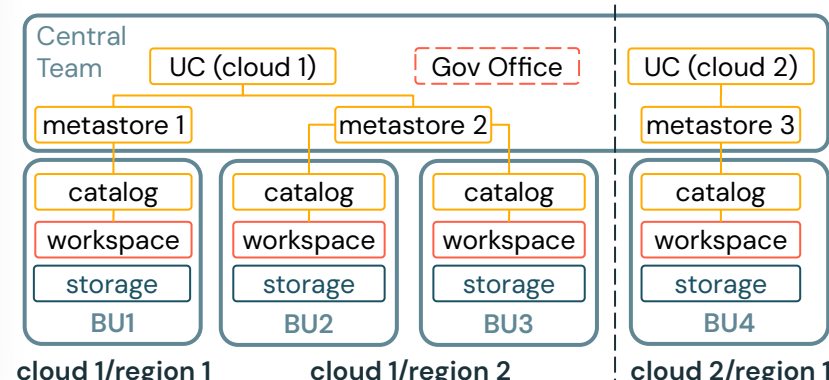
Multi-cloud and Multi-region

Centralized Governance



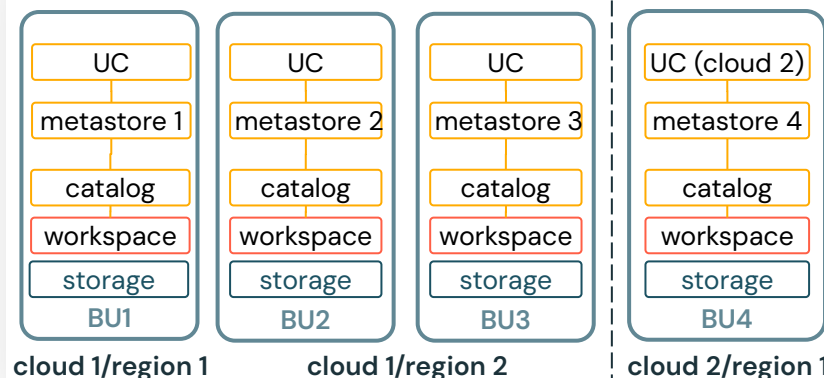
Same as single cloud, plus
– the central team will own second UC, metastores and catalogs

Federated Governance



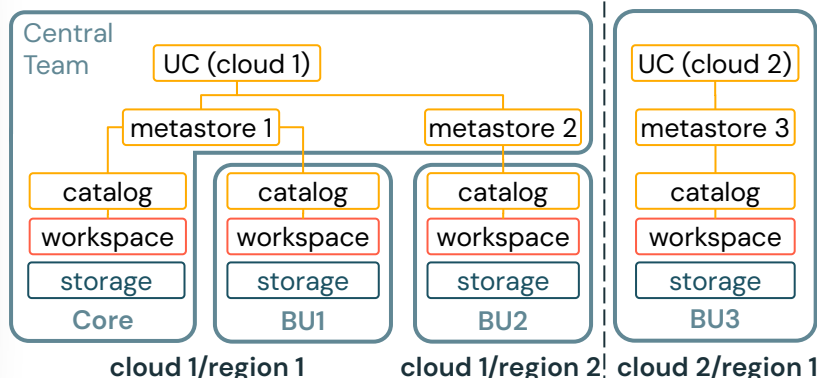
Same as single cloud, plus
– the central team will own the second Unity Catalog and metastores
– The BUs in the second cloud act like the BUs in the first cloud

Decentralized Governance



Same as single cloud

Hybrid Governance



Same as single cloud, plus
– the second cloud can be independent and another BU can own the second UC and metastores

Dependency for Lakehouse deployments

Different deployments demand for higher maturity and special models

